

# PATENT SPECIFICATION

## State-Adaptive Compliance Enforcement System with Multi-Spectrum Anomaly Detection and Failure-Mode Correction for Decentralised Finance Networks

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### 1. TECHNICAL FIELD

The present invention relates to systems and methods for enforcing anti-money laundering (AML) and regulatory compliance in decentralised finance (DeFi) transactions using blockchain smart contracts, machine learning risk scoring, and zero-knowledge cryptographic proofs. More particularly it relates to a compliance system that performs emphpre-settlement transaction flow control in a distributed ledger network by causing smart contracts to permit, delay (escrow), or reject execution of a transaction request based on: (i) a machine learning risk score; and (ii) a selected enforcement profile that deterministically defines numerical risk thresholds and cryptographic proof requirements to be verified on-chain as a prerequisite to settlement. In preferred embodiments the risk score is computed using (a) a multi-spectrum deviation representation that projects the transaction feature vector through multiple law-domain operators to form an attribution-rich anomaly tensor, and (b) a failure-mode correction layer that applies post-hoc correction operators to a frozen deployed model without retraining. In further embodiments the enforcement profile is selected in real time by a closed-form state-adaptive controller derived from a dynamical energy model of the transaction network.

### 2. BACKGROUND

#### 2.1. Technical Problems

DeFi protocols process financial transactions on blockchain networks without centralised intermediaries, creating four interacting technical problems that known art has not resolved in combination.

##### *2.1.1. Single-Perspective Anomaly Detection Fails to Cover the Full Space of Fraud Patterns*

Standard anomaly detectors (Isolation Forest, LOF, One-Class SVM, autoencoders) each produce a single scalar score. They reveal *that* a transaction is anomalous but not *which*

*mathematical aspect* makes it so. No single detector spans the full space of mathematical anomaly. Scores may concentrate in a narrow band, failing to discriminate among fraud subtypes (the coverage-AUC gap). No prior detector provides per-law-domain attribution required for downstream failure diagnosis.

### *2.1.2. Structural Blind Spots in Deployed Scoring Models*

Supervised models deployed for fraud detection fail to detect certain fraudulent transactions for structural reasons: mimicry of legitimate patterns (camouflage), informational sparsity (feature gap), novel activity scale (activity anomaly), and temporal distributional shift (temporal novelty). Fewer than 0.1% of blockchain transactions are labelled illicit, making retraining impractical. SHAP and LIME explain individual predictions in input-feature space but do not diagnose structural failure patterns or provide correction mechanisms without model retraining.

### *2.1.3. Pre-Settlement Enforcement Gap*

Blockchain analytics platforms (Chainalysis, Elliptic) operate retrospectively, identifying suspicious transactions after settlement when value transfer is irrecoverable. Protocol-level KYC/AML checks are bypassed when users transact directly through DeFi smart contracts. Privacy regulations (UK Data Protection Act 2018, EU GDPR) conflict with disclosure-intensive compliance models.

### *2.1.4. Static Enforcement Parameters*

Existing compliance systems apply fixed risk thresholds and fixed proof requirements regardless of the network's structural state. This is provably suboptimal: above a critical manifold in state space any constant enforcement configuration fails to achieve the state-dependent welfare optimum. No prior art derives a closed-form optimal enforcement function mapping a computed energy landscape to numerical thresholds and proof requirements.

## **2.2. Prior Art Limitations**

Known prior art includes: Isolation Forest, LOF, One-Class SVM (single-score, no attribution); deep autoencoders and ODIN (single latent representation); SHAP, LIME (feature-wise explanation, no structural correction); conformal prediction (calibrated sets, no failure diagnosis); blockchain analytics retrospective monitoring platforms; exchange-level KYC/AML not extending to protocol transactions; Basel III countercyclical capital buffer rule applying a fixed piecewise-linear function to credit-to-GDP gap with no closed-form state-dependent optimal policy; and agent-based financial network simulations lacking closed-form optimal policies and provable stability guarantees. None provides multi-law-domain anomaly tensor construction, four-component failure-mode correction on a frozen model, pre-settlement blockchain enforcement, and closed-form state-adaptive threshold and proof-requirement selection in combination.

### 3. SUMMARY OF THE INVENTION

The invention provides a unified state-adaptive compliance enforcement system for DeFi networks. The shared inventive concept is *controlling pre-settlement execution of decentralised finance transactions by selecting, from a predetermined set, an enforcement profile that deterministically defines numerical risk thresholds and on-chain cryptographic proof requirements, wherein the selection is computed from network instability metrics and the resulting policy is enforced by smart contracts before settlement*. Preferred embodiments further compute the risk score using a multi-spectrum anomaly tensor and correct structural blind spots of a frozen deployed model without retraining. The primary technical contributions are:

- (i) A **four-layer DeFi compliance architecture** (interface, compliance orchestration, offline training, infrastructure) enforcing policy actions before transaction settlement.
- (ii) A **multi-spectrum transaction anomaly engine** projecting each transaction feature vector through at least five independent mathematical law domains—statistical, dynamical, spectral, geometric, and exponential—to construct a multi-dimensional anomaly tensor  $T_{i,k,d}$ , with a formal coverage guarantee; and decomposing that tensor into magnitude, direction, and novelty components for per-law-domain attribution.
- (iii) A **boundary-centred dimension magnifier** rescaling each tensor dimension by its empirical class-separability, amplifying well-separated dimensions using a single global hyperparameter set ( $\gamma=5.0$ ,  $\lambda=10^{-4}$ ).
- (iv) A **weak-supervision offline training pipeline** comprising a composite teacher ensemble (autoencoder–gradient boosting, FATF rule-based, graph-structural) generating pseudo-labels from unlabelled blockchain data and an XGBoost student model trained on those labels.
- (v) A **failure-mode correction layer** operating on the frozen deployed student model: it decomposes each transaction into four near-orthogonal components—Camouflage ( $C$ ), Feature Gap ( $G$ ), Activity Anomaly ( $A$ ), Temporal Novelty ( $T$ )—computes a composite blind-spot score (MFLS) with self-calibrating weights, and applies a post-hoc correction operator to the student model’s predicted probability, producing a corrected risk score without model retraining.
- (vi) A **state-adaptive transaction flow control module** computing one or more network instability metrics from recent on-chain activity and selecting, from a predetermined set of enforcement profiles, a configuration deterministically defining: (a) a risk threshold profile ( $\tau_1, \tau_2, \tau_3, \tau_4$ ) for the five-tier decision matrix; and (b) a proof-requirement policy specifying, for at least one tier or transaction class, one or more cryptographic proofs required on-chain prior to settlement.
- (vii) A **gravitational interaction engine** modelling  $N$  financial institutions as particles under a pairwise erf-attraction plus logarithmic repulsion potential; a **spectral phase-transition detector** determining whether the network lies above the critical manifold; a **blind-spot energy module** computing a macro-level complementary

energy function from four deviation channels; and a **closed-form optimal adaptive policy**  $\gamma^*(X)$  derived from the Bellman equation that drives profile selection.

- (viii) A **zero-knowledge proof subsystem** configured to verify, on-chain, one or more privacy-preserving compliance predicates as prerequisites to settlement according to the proof-requirement policy, wherein the proofs are verifiable without revealing underlying personal data.
- (ix) **Smart contract infrastructure** enforcing all policy actions deterministically before settlement with an immutable audit trail.

## 4. DETAILED DESCRIPTION

### 4.1. System Architecture Overview

The system comprises four hierarchically organised layers.

**Layer I — Interface Layer.** Client SDKs (TypeScript, Python) and a RESTful API receive transaction requests, return risk scores, and surface compliance status. A web dashboard provides visual monitoring.

**Layer II — Compliance Orchestration Layer.**

- *Multi-spectrum transaction anomaly engine* (§4.3)
- *Failure-mode correction layer* (§4.4)
- *Graph analytics module*: transaction graph construction; centrality, PageRank, community detection.
- *Sanctions screening module*: OFAC and EU database checks.
- *State-adaptive transaction flow control module* (§4.7).
- *Deterministic policy engine*: maps corrected scores to policy actions via the thresholds of the selected enforcement profile.

**Layer III — Offline Training Layer.** Composite teacher ensemble, pseudo-label generation, student model training, cross-validation, sigmoid calibration.

**Layer IV — Infrastructure Layer.** Blockchain smart contracts, privacy-preserving proof circuits, databases, distributed file storage.

### 4.2. Pre-Settlement Smart Contract Enforcement Workflow

In a preferred embodiment the system enforces compliance by controlling execution of a transaction request at the point of settlement in the blockchain transaction processing network. A representative workflow is:

- (a) A client application submits a transaction request to a core smart contract function, the request comprising at least a sender address, a recipient address, and a transaction value.
- (b) The compliance orchestration layer computes a risk score from the feature vector and selects an enforcement profile based on instability metrics computed over a recent sliding time window of on-chain activity.

- (c) The selected enforcement profile deterministically defines (i) a set of numerical thresholds for the compliance decision matrix, and (ii) a proof-requirement policy specifying which cryptographic proofs must be verified for one or more threshold tiers or transaction classes.
- (d) If the proof-requirement policy requires one or more proofs, the transaction is prevented from settling unless verification succeeds on-chain. In a preferred embodiment the proof is verified by calling a verifier contract that records a verification result, a proof identifier, and a timestamp, and the core contract checks the verification result and that the verification is fresh within a predetermined validity window and has not been replayed.
- (e) Based on the risk score and the selected thresholds, the core contract either (i) authorises execution, (ii) transfers value into an escrow contract (time-lock), or (iii) rejects execution by reverting.

This workflow provides a technical effect beyond risk scoring *per se*: it prevents irreversible settlement of non-compliant transactions by enforcing deterministic preconditions within the transaction processing network itself, and it enforces privacy-preserving proof-gating without requiring disclosure of underlying identity or score values.

### 4.3. Multi-Spectrum Transaction Anomaly Engine (UDL)

#### 4.3.1. Universal Deviation Principle

**Definition (Deviation-Separating Family):** An operator family  $\{\Phi_k\}_{k=1}^K$  is *deviation-separating* if the stacked Jacobian  $J(x) = [D\Phi_1(x)^\top \cdots D\Phi_K(x)^\top]^\top$  has full column rank for all  $x$  in the transaction feature domain.

**Theorem 1 (Coverage Guarantee):** If  $\{\Phi_k\}$  is deviation-separating, no anomalous transaction can simultaneously match the normal projection under all operators. This provides a detection guarantee absent from every single-perspective detector.

**Theorem 2 (Global Bound):** On a compact anomaly set a quantitative lower bound on minimum deviation across all operators can be established.

**Theorem 3 (Finite-Sample Guarantee):** Given  $n$  normal training samples the coverage guarantee holds with probability  $\geq 1 - \delta$  for a sample complexity depending on the operator family's covering number.

**Theorem 4 (Information-Optimal Fusion):** Optimal operator-output weights are the solution to a semidefinite programme.

**Theorem 5 (Greedy Operator Selection):** A greedy algorithm achieves a  $(1 - 1/e)$  approximation to the optimal operator subset under a submodular information criterion.

#### 4.3.2. Spectrum Operators

Five spectrum operators  $\Phi_k$  are instantiated on the 167-dimensional transaction feature vector:

**Statistical Operator ( $\Phi_{\text{stat}}$ ).** Shannon entropy of the local neighbourhood; KL divergence from the reference distribution; Hellinger distance; skewness; kurtosis.

**Dynamical Operator** ( $\Phi_{\text{chaos}}$ ). Lyapunov exponent from the temporal transaction context; recurrence rate; approximate entropy.

**Spectral Operator** ( $\Phi_{\text{freq}}$ ). Power spectral density divergence; low-to-high frequency energy ratio; spectral entropy; spectral centroid shift.

**Geometric Operator** ( $\Phi_{\text{geom}}$ ). Mahalanobis distance from the reference centroid; cosine similarity to the reference direction; norm ratio.

**Exponential Operator** ( $\Phi_{\text{exp}}$ ).

$$\Phi_{\text{exp}}(x) = \exp\left(\alpha \cdot \frac{x-\mu}{\sigma}\right) \quad (1)$$

amplifying small deviations invisible under linear representations.  $\mu$ ,  $\sigma$  are reference statistics computed on normal transactions.

In an enhanced embodiment up to 14 composable operators are available (phase-space, topological, kernel, rank, wavelet). An operator correlation audit confirms at most one pair exceeds the redundancy threshold ( $\rho < 0.85$ ).

#### 4.3.3. Anomaly Tensor Construction

Each transaction feature vector  $x_i$  is projected through all  $K$  operators to yield a three-dimensional anomaly tensor:

$$T_{i,k,d} \quad \text{for transaction } i, \text{ operator } k, \text{ dimension } d \quad (2)$$

#### 4.3.4. Magnitude–Direction–Novelty Decomposition

- (a) **Magnitude**  $r_i = \|T_i\|$ : total deviation amount.
- (b) **Direction**  $\hat{D}_i = T_i/\|T_i\|$ : unit tensor indicating which law domains drive the anomaly.
- (c) **Novelty**  $\theta_i$ : angular distance of  $\hat{D}_i$  from reference deviation directions in training data.

#### 4.3.5. Boundary-Centred Dimension Magnifier

Each tensor dimension  $d$  is rescaled:

$$r'_d = \tanh\left(k_d \cdot \frac{r_d - b_d}{\sigma_d}\right) \quad (3)$$

where  $b_d$  is the decision boundary in dimension  $d$ ,  $\sigma_d$  is the local scale, and  $k_d = \gamma \cdot \delta_d$  is the per-dimension steepness. The discriminability score  $\delta_d$  is computed from three complementary measures: Cohen's  $d$  (effect size), class-distribution overlap area, and per-dimension AUC. The global parameters  $\gamma=5.0$  and QDA regularisation  $\lambda=10^{-4}$  constitute the single hyperparameter set used across all deployments.

#### 4.3.6. Classification and Coverage

The rescaled tensor is classified by QDA-Magnified (preferred) or a hybrid auto-selector that evaluates multiple strategies and maximises coverage:

$$\text{Coverage@}k = \frac{|\{x : \text{rank}(x) \leq k, x \in \text{anomaly}\}|}{|\text{anomaly}|} \quad (4)$$

In the preferred embodiment mean coverage exceeds 91%. The anomaly tensor is the primary input to both the failure-mode correction layer (§4.4) and the risk scoring engine.

### 4.4. Failure-Mode Correction Layer (BSDT)

The failure-mode correction layer receives the anomaly tensor from §4.3 and the predicted probability  $p(x)$  from the deployed frozen student model, and produces a corrected probability  $p^*(x)$  without modifying the model's parameters.

#### 4.4.1. Relationship to Spectrum Operators

The four BSDT components correspond to specialised instantiations of the UDL spectrum operators applied to the student model's feature space: Camouflage ( $C$ ) corresponds to the geometric operator applied to the legitimate-class centroid; Feature Gap ( $G$ ) corresponds to the statistical operator measuring informational sparsity; Activity Anomaly ( $A$ ) corresponds to the dynamical operator measuring deviation from known fraud activity; Temporal Novelty ( $T$ ) corresponds to the spectral and dynamical operators applied over the temporal axis of the training distribution. At the macro level the same four channel types reappear in the blind-spot energy module of the systemic risk controller (§4.7).

#### 4.4.2. Camouflage Component

$$C(x) = 1 - \frac{\|x - \mu_{\text{legit}}\|_2}{d_{\text{max}}} \quad (5)$$

$C(x) \approx 1$  indicates that the transaction mimics legitimate patterns, depressing the student model's suspicion score.

#### 4.4.3. Feature Gap Component

$$G(x) = \frac{|\{j : |x_j| < \varepsilon\}|}{d}, \quad \varepsilon = 10^{-8} \quad (6)$$

High  $G(x)$  indicates many uninformative features, depriving the model of discriminative signal.

#### 4.4.4. Activity Anomaly Component

$$A(x) = \sigma \left( \frac{\log(1 + |\text{activity}(x)|) - \mu_{\text{ref}}}{\sigma_{\text{ref}}} \right) \quad (7)$$

Measures deviation of transaction activity from the reference detected-fraud population; captures fraud operating at scales not represented in training labels.

#### 4.4.5. Temporal Novelty Component

$$T(x) = \sigma\left(\frac{1}{2}(\bar{m}(x) - 2)\right) \quad (8)$$

$\bar{m}(x)$  is the mean temporal distance to the  $k$  nearest training neighbours, normalised by the temporal span.

#### 4.4.6. Composite Blind-Spot Score and Self-Calibrating Weights

$$\text{MFLS}(x) = \sum_{i \in \{C, G, A, T\}} w_i \cdot S_i(x) \quad (9)$$

The weights  $w_i$  (normalised to sum to 1) are computed by one or more of: mutual information; Fisher variance ratio; Bayesian online updating; gradient-based optimisation. The four components are verified near-orthogonal: mean NMI  $\in [0.051, 0.064]$  across all pairs (threshold 0.30); all VIF  $< 1.45$ .

#### 4.4.7. Correction Operators

Four post-hoc correction operators adjust  $p(x)$  based on the MFLS:

**Additive.**

$$p^*(x) = p(x) + \lambda \cdot \text{MFLS}(x) \cdot (1 - p(x)) \cdot \mathbb{I}[p(x) < \tau] \quad (10)$$

**SignedLR.**

$$p^*(x) = \text{clip}(p(x) + \beta_0 + \sum_i \beta_i S_i(x), 0, 1) \quad (11)$$

**QuadSurf (preferred).**

$$p^*(x) = \text{clip}\left(p(x) + \phi_2([C, G, A, T])^\top \hat{\beta}, 0, 1\right) \quad (12)$$

$\phi_2$  is the degree-2 polynomial feature map;  $\hat{\beta}$  are ridge-regression coefficients ( $\alpha=1.0$ ).

**ExpGate.**

$$p^*(x) = \text{clip}(p(x) + g(x) \cdot q(x), 0, 1) \quad (13)$$

$q(x)$  is the QuadSurf output;  $g(x) = \sigma(v^\top [C, G, A, T] + b)$  is a learned sigmoid gate.

The corrected probability  $p^*(x)$  is multiplied by 1000 to yield the integer risk score used by the decision matrix.

### 4.5. Weak Supervision Offline Training Pipeline

Three independent teacher models generate pseudo-labels from unlabelled blockchain transaction data:

- (a) **Autoencoder–Gradient Boosting Teacher** ( $s_{\text{AE-XGB}}$ ): autoencoder reconstruction error and latent features feed an XGBoost model.
- (b) **FATF Rule-Based Teacher** ( $s_{\text{FATF}}$ ): encodes FATF red-flag indicators as deterministic scoring rules.



- (c) **Graph-Based Teacher** ( $s_{\text{Graph}}$ ): suspicion scores from proximity to high-risk addresses, temporal clustering, and unusual connectivity.

Composite teacher score:

$$s_{\text{teacher}}(x) = w_1 s_{\text{AE-XGB}}(x) + w_2 s_{\text{FATF}}(x) + w_3 s_{\text{Graph}}(x) \quad (14)$$

Preferred:  $w_1=0.4$ ,  $w_2=0.3$ ,  $w_3=0.3$ . Binary pseudo-labels train an XGBoost student model on 167 engineered features. Raw output is sigmoid-recalibrated:

$$\hat{p}_{\text{cal}}(x) = \sigma(k \cdot (s(x) - P_{75})) \quad (15)$$

where  $k = 30$  and  $P_{75}$  is the 75th percentile of the training score distribution. The BSDT correction layer (§4.4) then produces the final corrected integer risk score.

#### 4.6. Compliance Decision Matrix

The deterministic policy engine maps integer scores  $[0, 1000]$  to policy actions according to thresholds drawn from the active enforcement profile:

| Score Range        | Policy Action     | Smart Contract Effect      |
|--------------------|-------------------|----------------------------|
| $[0, \tau_1)$      | APPROVE           | Immediate execution        |
| $[\tau_1, \tau_2)$ | APPROVE + MONITOR | Execution; monitoring flag |
| $[\tau_2, \tau_3)$ | REVIEW            | Queued for manual review   |
| $[\tau_3, \tau_4)$ | ESCROW            | Time-locked escrow         |
| $[\tau_4, 1000]$   | BLOCK             | On-chain rejection         |

Under the stable profile:  $(\tau_1, \tau_2, \tau_3, \tau_4) = (200, 400, 600, 800)$ . The state-adaptive module (§4.7) selects profiles that adjust these thresholds and impose additional proof requirements.

##### 4.6.1. Example Enforcement Profiles

In a preferred embodiment the predetermined set of enforcement profiles includes at least a *stable* profile, a *stressed* profile, and a *critical* profile. Each profile deterministically defines both a threshold tuple  $(\tau_1, \tau_2, \tau_3, \tau_4)$  and a proof-requirement policy.

| Profile  | $(\tau_1, \tau_2, \tau_3, \tau_4)$ | Proof-gated tiers | Required proofs                               |
|----------|------------------------------------|-------------------|-----------------------------------------------|
| Stable   | (200, 400, 600, 800)               | $[\tau_3, 1000]$  | non-membership predicate                      |
| Stressed | (250, 450, 650, 850)               | $[\tau_2, 1000]$  | non-membership + range predicate              |
| Critical | (300, 500, 700, 900)               | $[\tau_1, 1000]$  | non-membership + range + credential predicate |

The example profiles are illustrative and may be modified for a given deployment. For a fixed system state, the selected profile and the resulting on-chain decisions are deterministic and reproducible.

## 4.7. State-Adaptive Transaction Flow Control Module

### 4.7.1. Instability Metrics

Network instability metrics are computed from:

- (a) spectral radius or centrality concentration of the transaction graph over a sliding time window;
- (b) a temporal burst indicator from the transaction count process; and/or
- (c) a shift of the corrected risk score distribution relative to its reference distribution.

### 4.7.2. Enforcement Profile Selection

The module selects one profile from the discrete set  $\{stable, stressed, critical\}$ . Each profile deterministically defines:

- (a) a risk threshold profile  $(\tau_1, \tau_2, \tau_3, \tau_4)$ ; and
- (b) a proof-requirement policy specifying which privacy-preserving compliance predicate proofs must be verified on-chain for at least one tier or transaction class, the compliance predicates comprising at least one of: (i) a set non-membership predicate, (ii) a numerical range predicate, or (iii) a credential validity predicate.

### 4.7.3. Gravitational Interaction Engine

When multi-institution financial metrics are available, the module models  $N$  institutions as particles in  $\mathbb{R}^d$ :

$$\dot{X} = -\alpha(X - \mu \mathbf{1}^\top) - \nabla_X \Phi_{\text{pair}}(X) - \sum_k \beta_k D\Phi_k(X)^\top D_k(X) \quad (16)$$

### 4.7.4. Pairwise Interaction Potential

$$\phi(r) = \gamma \cdot \frac{\sigma\sqrt{\pi}}{2} \text{erf}(r/\sigma) - \gamma\lambda_{\text{rep}} \cdot \log(r + \varepsilon) \quad (17)$$

Preferred parameters:  $\alpha=0.10$ ,  $\gamma=1.00$ ,  $\sigma=1.00$ ,  $\lambda_{\text{rep}}=0.10$ ,  $\varepsilon=10^{-5}$ .

### 4.7.5. Macro-Level Blind-Spot Energy Module

A macro-level energy function is computed from four deviation channels that mirror, at the network level, the BSDT components at the transaction level:

- (a) enstrophy anomaly  $\delta_C$ : rotational energy deviation (analogous to Camouflage);
- (b) spectral gap  $\delta_G$ : structural fragility indicator (analogous to Feature Gap);
- (c) alignment anomaly  $\delta_A$ : co-movement deviation (analogous to Activity Anomaly); and
- (d) temporal novelty  $\delta_T$ : drift from reference state patterns (analogous to Temporal Novelty).

$$E_{\text{BS}}(X) = \|(X - \mu_0)^\top \Sigma_0^{-1} (X - \mu_0)\|_F \quad (18)$$

$\mu_0$  and  $\Sigma_0$  are reference mean and covariance estimated by Ledoit–Wolf Oracle shrinkage.

#### 4.7.6. Critical Manifold

**Theorem (Spectral Phase Transition):**

$$\mathcal{C} = \{X : \lambda_{\max}(\rho(X) \cdot \ell(X) \cdot W) = 1\} \quad (19)$$

Above  $\mathcal{C}$  state-adaptive friction is both necessary and sufficient for welfare optimality.

#### 4.7.7. Closed-Form Optimal Adaptive Policy

**Theorem (Optimal Friction):** The Bellman-optimal friction coefficient is:

$$\gamma^*(X) \approx \frac{\beta \|\nabla E\|^2}{2\kappa + \beta \|\nabla E\|^2} \cdot \frac{E(X)}{E(X) + \theta} \quad (20)$$

$\gamma^* \rightarrow 0$  as  $E \rightarrow 0$  (minimum friction when stable);  $\gamma^* \rightarrow 1$  as  $E \rightarrow \infty$  (maximum friction under severe instability). The adaptive damping factor  $E/(E + \theta)$  is the same functional form as the BSDT damping function  $\gamma^*(E) = E/(E + \theta)$  applied at the transaction level by the correction operators, linking both levels of the system through a single analytical form.

Profile selection: below  $\gamma_1$  — *stable*; between  $\gamma_1$  and  $\gamma_2$  — *stressed*; above  $\gamma_2$  — *critical*.

#### 4.7.8. Navier–Stokes Analogy

Institution position changes  $\dot{X}$  are interpreted as a velocity field decomposed into strain  $S$  and vorticity  $\Omega$ . Enstrophy  $\mathcal{E} = \frac{1}{2}\|\omega\|_F^2$  quantifies rotational energy. The Beale–Kato–Majda criterion adapted to financial networks serves as a singularity indicator. The adaptive friction coefficient is applied as an adaptive viscosity  $\nu = \nu_0(1 + \gamma^*(E_{BS}))$ . Minsky regime classification (hedge, speculative, Ponzi) emerges from the cosine alignment between the gravitational and blind-spot gradients.

### 4.8. Zero-Knowledge Proof Subsystem (zkNAF)

In preferred embodiments, the system includes one or more zero-knowledge proof circuits implementing privacy-preserving compliance predicates. In one embodiment the proof system is a succinct non-interactive argument of knowledge (SNARK), for example Groth16.

**Example: Set Non-Membership Predicate.** The prover shows an address is absent from a cryptographic set commitment without revealing the address.

**Example: Numerical Range Predicate.** The prover demonstrates  $s_{\min} \leq s(x) \leq s_{\max}$  without revealing the exact corrected score.

**Example: Credential Validity Predicate.** The prover demonstrates satisfaction of an identity or credential status with an accredited provider without revealing identity data.

Each proof may carry a predetermined validity interval. An on-chain verifier contract performs verification and records the result together with a timestamp and a proof identifier.

#### 4.9. Smart Contract Architecture

- **Core Contract:** on-chain decision matrix; immutable audit record.
- **Escrow Contract:** time-locked escrow; release and dispute escalation.
- **ZK Verifier Contract:** verification of selected zero-knowledge proofs; binary result stored with timestamp and proof identifier.
- **Cross-Chain Bridge Contract:** requires compliance verification on both source and destination chains.
- **Dispute Resolution Contract:** structured escalation for escrowed disputes.

All enforcement decisions execute before the corresponding transaction settles on the blockchain network.

#### 4.10. Performance Characteristics

**DeFi Compliance System** (2.64 million Ethereum transactions):

| Metric                             | Value               |
|------------------------------------|---------------------|
| Meta-Ensemble ROC-AUC (test)       | 1.0000              |
| Meta-Ensemble ROC-AUC (5-fold CV)  | $0.9965 \pm 0.0049$ |
| Meta-Ensemble PR-AUC               | 0.9999              |
| Transaction XGBoost AUC            | 0.9878    F1: 0.901 |
| Cross-domain mean AUC (9 datasets) | 0.9962              |
| Escrow contract gas cost           | 287,890 gas         |

**UDL Anomaly Engine** (single hyperparameter set,  $\gamma=5.0$ ):

| Dataset             | QDA-Magnified AUC | Hybrid AUC |
|---------------------|-------------------|------------|
| Synthetic           | 1.000             | 0.999      |
| Mimic (adversarial) | 1.000             | 0.999      |
| Mammography         | 0.947             | 0.911      |
| Shuttle             | 0.989             | 0.991      |
| Pendigits           | 0.777             | 0.960      |
| Mean                | 0.943             | 0.972      |

**BSDT Correction Layer** (792 variant–model–dataset combinations):

| Metric                                | Value                              |
|---------------------------------------|------------------------------------|
| Best variant (QuadSurf) mean F1       | 0.787 (+43% relative)              |
| False positive reduction              | 88.1% (44,740 $\rightarrow$ 5,345) |
| Maximum recall recovery               | +84.4 percentage points            |
| Shuttle rescue F1                     | 0.134 $\rightarrow$ 0.968          |
| Cross-domain credit card MFLS AUC     | 0.885                              |
| Cross-domain aerospace MFLS AUC       | 0.982                              |
| Cross-domain medical imaging MFLS AUC | 0.916                              |

**Adaptive Friction Controller** (FDIC SDI data, 140 quarters):

| Configuration                | $N$ | AUROC       | GFC Lead   |
|------------------------------|-----|-------------|------------|
| Institution-type (7 classes) | 7   | $\geq 0.70$ | 6 quarters |
| Bank-level (individual)      | 30  | 0.805       | 6 quarters |
| G-SIB cross-border           | 20  | 0.781       | 6 quarters |

Super-criticality: 68.6% of sample above  $\mathcal{C}$ . GFC welfare loss of inaction: 19.57 pp consumption-equivalent.

## 5. CLAIMS

*The following claims define the scope of protection sought. Features of any dependent claim may be combined with any independent claim or with other dependent claims.*

1. A computer-implemented transaction compliance system for a decentralised finance transaction processing network, the system comprising:
  - (a) an interface layer configured to receive, from a client application, a transaction request comprising at least a sender address, a recipient address, and a transaction value;
  - (b) a risk scoring engine configured to compute a numerical risk score for the transaction request by:
    - (i) constructing a feature vector comprising address-level, transaction-level, and graph-level features;
    - (ii) projecting the feature vector through a plurality of spectrum operators each computing deviation measurements under a different mathematical law domain to construct a multi-dimensional anomaly tensor;
    - (iii) applying a trained machine learning model to the anomaly tensor to compute the numerical risk score;
  - (c) a state-adaptive transaction flow control module configured to compute one or more network instability metrics from recent transaction activity and to compute a state-dependent friction coefficient  $\gamma^*(X)$  based on the one or more network instability metrics, and to select, from a predetermined set of enforcement profiles, an enforcement configuration based on the computed friction coefficient  $\gamma^*(X)$ , the enforcement configuration comprising:
    - (i) a risk threshold profile defining numerical thresholds for a deterministic compliance decision matrix; and
    - (ii) a proof-requirement policy defining, for at least one threshold tier or transaction class, one or more cryptographic proofs required to be verified on-chain as a prerequisite to settlement;
  - (d) a deterministic policy engine configured to map the risk score to a policy action according to the selected risk threshold profile and to enforce the proof-requirement policy by causing a settlement-preventing action when required proofs are absent, expired, or fail on-chain verification, wherein the settlement-preventing action comprises at least one of: causing a core smart contract function to revert execution of a transaction call, or preventing release of value from an escrow smart contract; and
  - (e) smart contract infrastructure deployed on a blockchain network and configured to implement, before settlement, the policy action determined by the deterministic policy engine, including a core contract configured to reject or authorise execution, an escrow contract configured to time-lock value transfers for an escrow policy action, and a zero-knowledge proof verifier contract configured to verify said cryptographic proofs, wherein the core contract is further configured to require, for a transaction class or threshold tier specified by the proof-requirement policy, a successful verification result from the zero-knowledge proof verifier contract prior to permitting execution;

wherein the smart contract infrastructure prevents settlement of the transaction request unless the policy action permits execution and, when required by the proof-requirement policy, the cryptographic proofs are verified on-chain.

2. A computer-implemented method of enforcing compliance in decentralised finance transactions on a blockchain network, the method comprising:
  - (a) receiving a transaction request comprising at least a sender address, a recipient address, and a transaction value;
  - (b) computing a numerical risk score by:
    - (i) constructing a feature vector comprising address-level, transaction-level, and graph-level features;
    - (ii) projecting the feature vector through a plurality of spectrum operators each computing deviation measurements under a different mathematical law domain to construct a multi-dimensional anomaly tensor;
    - (iii) applying a trained machine learning model to the anomaly tensor to compute the numerical risk score;
  - (c) computing one or more network instability metrics from recent transaction activity, computing a state-dependent friction coefficient  $\gamma^*(X)$  based on the one or more network instability metrics, and selecting an enforcement profile from a predetermined set based on the computed friction coefficient  $\gamma^*(X)$ , the selected profile deterministically defining:
    - (i) a risk threshold profile for a deterministic decision matrix; and
    - (ii) a proof-requirement policy specifying one or more cryptographic proofs to be verified on-chain as a prerequisite to settlement for at least one threshold tier or transaction class;
  - (d) mapping the risk score to a policy action using the decision matrix with thresholds from the selected risk threshold profile;
  - (e) before settlement, enforcing the proof-requirement policy by verifying the specified cryptographic proofs on-chain using a zero-knowledge proof verifier smart contract and preventing execution of the transaction when required proofs are absent, expired, or fail verification, wherein preventing execution comprises reverting a core smart contract function call or preventing release of value from an escrow smart contract; and
  - (f) executing a corresponding smart contract function on a blockchain network to implement the policy action, including at least one of: authorising execution, transferring value into escrow, maintaining value in escrow until a release condition is satisfied, or rejecting execution by reversion; and recording the outcome on-chain as an immutable audit record.
3. A system according to claim 1 or a method according to claim 2, wherein computing the risk score comprises projecting the feature vector through a plurality of spectrum operators each computing deviation measurements under a different mathematical law domain to construct a multi-dimensional anomaly tensor, and using the anomaly tensor as an input to the trained machine learning model.

4. A system or method according to claim 3, wherein the plurality of spectrum operators comprises at least: a statistical operator measuring distributional deviations, a dynamical operator measuring temporal dynamical deviations, a spectral operator measuring frequency-domain deviations, a geometric operator measuring geometric relationship deviations, and an exponential operator measuring deviations under exponential amplification.
5. A system or method according to claim 3 or claim 4, wherein the anomaly tensor is decomposed into magnitude, direction, and novelty components, and the direction and novelty components provide per-law-domain attribution identifying which law domains contribute to a risk assessment.
6. A system or method according to any of claims 3 to 5, wherein the spectrum operators collectively satisfy a deviation-separating condition such that the stacked Jacobian of the operators has full column rank over the transaction feature domain, providing a coverage guarantee that no anomalous transaction can simultaneously match normal data across all operators.
7. A system or method according to any of claims 1 to 6, wherein the boundary-centred dimension magnifier rescales each anomaly tensor dimension by:

$$r'_d = \tanh\left(k_d \cdot \frac{r_d - b_d}{\sigma_d}\right)$$

where  $k_d = \gamma \cdot \delta_d$ ,  $\delta_d$  is a per-dimension discriminability score computed from Cohen's  $d$ , class-distribution overlap, and per-dimension AUC, and  $\gamma$  is a single global steepness parameter.

8. A system according to claim 1 or a method according to claim 2, wherein computing the risk score further comprises computing, from an anomaly tensor and a predicted probability output by a frozen deployed student model, four near-orthogonal failure-mode components comprising camouflage, feature gap, activity anomaly and temporal novelty, aggregating them into a composite blind-spot score, and applying a correction operator to the predicted probability to produce a corrected risk score without retraining or updating parameters of the trained machine learning model.
9. A system or method according to claim 8, wherein the correction operator is a QuadSurf correction:

$$p^*(x) = \text{clip}\left(p(x) + \phi_2([C, G, A, T])^\top \hat{\beta}, 0, 1\right)$$

where  $\phi_2$  is the degree-2 polynomial feature map and  $\hat{\beta}$  are ridge-regression coefficients.

10. A system or method according to claim 8, wherein the correction operator is an exponential gating correction comprising a QuadSurf output modulated by a learned sigmoid gate  $g(x) = \sigma(v^\top [C, G, A, T] + b)$ .
11. A system or method according to any of claims 8 to 10, wherein the four failure-mode components are aggregated as:

$$\text{MFLS}(x) = \sum_{i \in \{C, G, A, T\}} w_i \cdot S_i(x)$$



and the weights  $w_i$  are automatically computed using one or more of mutual information, Fisher variance ratio, Bayesian online updating, or gradient-based optimisation.

12. A system or method according to claim 1 or claim 2, wherein the enforcement profile is selected from a discrete set and the selection is determined by whether the computed friction coefficient  $\gamma^*(X)$  lies between predetermined lower and upper thresholds.
13. A system or method according to claim 1 or claim 2, wherein the proof-requirement policy comprises, for at least one enforcement profile, requiring at least one privacy-preserving cryptographic proof establishing satisfaction of a compliance predicate, wherein the compliance predicate comprises at least one of: (i) a set non-membership predicate, (ii) a numerical range predicate, or (iii) a credential validity predicate.
14. A system or method according to claim 13, wherein the verifier smart contract records, for a successful proof verification, at least a proof identifier and a verification timestamp, and the core smart contract prevents settlement when the proof identifier has been previously used or when the verification timestamp is older than a predetermined validity threshold.
15. A system according to claim 1 or a method according to claim 2, wherein the proof-requirement policy deterministically defines one or more proof types as a function of (i) the computed state-dependent friction coefficient  $\gamma^*(X)$  and (ii) a per-law-domain attribution derived from the anomaly tensor, such that at least one of: (a)  $\gamma^*(X)$  exceeding a friction threshold or (b) an attribution component exceeding an attribution threshold causes the deterministic policy engine to require at least one of: a set non-membership predicate proof, a numerical range predicate proof, or a credential validity predicate proof.
16. A system or method according to claim 1 or claim 2, wherein the state-dependent friction coefficient is:

$$\gamma^*(X) \approx \frac{\beta \|\nabla E\|^2}{2\kappa + \beta \|\nabla E\|^2} \cdot \frac{E(X)}{E(X) + \theta}$$

where  $E(X)$  is the combined systemic energy,  $\kappa$  is an economic cost parameter,  $\beta$  is a systemic damage coefficient, and  $\theta = \kappa/\beta$ ; and wherein the adaptive damping factor  $E(X)/(E(X) + \theta)$  is the same functional form applied by the failure-mode correction module at the transaction level.

17. A system or method according to claim 16, wherein a state-adaptive transaction flow control module comprises a gravitational interaction engine configured to model financial institutions as particles interacting through a pairwise potential, a spectral monitoring module configured to determine a phase-transition condition, and an adaptive policy module configured to compute the friction coefficient.
18. A system or method according to claim 1 or claim 2, wherein selecting the enforcement profile further deterministically defines at least one of: a transaction rate limit, a queue delay, an escrow duration, an escrow release condition window, or a maximum transaction value limit.

19. A system according to claim 1, further comprising a cross-chain bridge contract requiring compliance verification on both a source chain and a destination chain before releasing bridged funds.
20. A system according to claim 1 or a method according to claim 2, wherein computing the one or more network instability metrics comprises constructing, over a sliding time window of on-chain activity, a transaction graph and computing at least one of: (i) a spectral radius of an adjacency or interaction matrix derived from the transaction graph, (ii) a centrality concentration metric, or (iii) a temporal burst metric from a transaction count process.
21. A system according to claim 1 or a method according to claim 2, wherein computing the one or more network instability metrics comprises modelling, over the sliding time window, aggregated transaction flow as a fluid-like flow field and computing at least one of: a divergence metric, a vorticity metric, or an effective viscosity parameter indicative of liquidity friction, and selecting the enforcement profile based on the at least one computed metric.
22. A system according to claim 1 or a method according to claim 2, wherein computing the one or more network instability metrics comprises classifying at least one entity, address cluster, or counterparty into a Minsky regime comprising at least one of: hedge, speculative, or Ponzi, based on refinancing dependence inferred from on-chain behaviour, and selecting the enforcement profile based on the classification.
23. A system according to claim 1 or a method according to claim 2, wherein the predetermined set of enforcement profiles is stored in an on-chain data structure of the core contract as a mapping from a profile identifier to (i) the risk threshold profile and (ii) the proof-requirement policy, and wherein selecting the enforcement profile comprises selecting a said profile identifier.
24. A system according to claim 1 or a method according to claim 2, wherein enforcing the proof-requirement policy comprises: (i) calling a zero-knowledge proof verifier contract that stores a verification result, a proof identifier, and a verification timestamp, and (ii) checking in the core contract that the stored verification result indicates success, that the proof identifier has not been previously used, and that the verification timestamp is within a predetermined validity window prior to permitting execution or release from escrow.
25. A system according to claim 1, wherein the core contract emits an on-chain event log comprising at least: a transaction identifier, a selected profile identifier, a policy action identifier, and a proof-verification status, thereby providing an immutable audit trail of pre-settlement enforcement.

## ABSTRACT

A computer-implemented system controls pre-settlement execution of DeFi transactions on a blockchain network. A risk scoring engine constructs an anomaly tensor by applying multiple spectrum operators to transaction features and applies a trained model to compute a risk score. A state-adaptive module computes network instability metrics and a state-dependent friction coefficient  $\gamma^*(X)$  and selects an enforcement profile that sets risk thresholds and a proof-requirement policy. Smart contracts enforce the profile before settlement by reverting execution or time-locking value in escrow unless required zero-knowledge proofs are verified on-chain. A verifier contract stores verification results with timestamps and proof identifiers and the core contract checks freshness within a predetermined validity window and prevents replay. On-chain events record the selected profile and outcome as an immutable audit trail.